

# From features to slice: parameter-cloud modeling of spatial transcriptomics for simulation and 3D interpolatory augmentation

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**Abstract.** Computational simulation and data augmentation of spatial transcriptomics (ST) are essential for quantitative benchmarking, reproducibility, and methodological innovation. Yet current models often lack flexibility in controlling spatial and transcriptional heterogeneity, fail to capture higher-order gene dependencies, and rarely extend to three-dimensional or alignment-aware contexts. Here we present FEAST, a computational infrastructure that models ST data within a parameter cloud - a latent manifold encoding gene-level mean, variance, and sparsity. By sampling and perturbing this manifold, FEAST generates high-fidelity synthetic slices with tunable spatial and transcriptional variation, enabling systematic evaluation of clustering, deconvolution, and spatial alignment algorithms. Beyond two dimensions, FEAST performs 3D parameter-cloud interpolation guided by optimal transport and benchmark alignment, reconstructing continuous tissue architectures while preserving molecular coherence. Together, these capabilities establish FEAST as a foundational platform for standardized benchmarking, data augmentation, and 3D reconstruction in spatial transcriptomics. The source code and tutorials for FEAST are publicly available at <https://github.com/maiziezhoulab/FEAST>, and can be installed via Pypi at <https://pypi.org/project/FEAST-py/>.

**Keywords:** Spatial Transcriptomics · ST Simulation · Computational Interpolation · 3D Reconstruction

# 1 Introduction

Spatial transcriptomics (ST) technologies have revolutionized transcriptomic profiling by capturing gene expression within its native tissue context.<sup>1–7</sup> By combining molecular readouts with spatial localization, ST enables gene expression maps that reveal the organization of transcriptional programs across tissue architecture, providing new insights into biological topics like development, neural circuitry, and disease. Alongside these experimental breakthroughs, a rapidly expanding ecosystem of computational methods has emerged to analyze ST data, encompassing spatial domain identification,<sup>8–11</sup> cell-type deconvolution,<sup>12</sup> multi-slice alignment and integration,<sup>13–15</sup> and three-dimensional (3D) tissue reconstruction.<sup>16,17</sup>

As these analytical frameworks proliferate, the lack of standardized, controllable, and interpretable simulation benchmarks has become a critical bottleneck for rigorous validation and reproducibility.<sup>18</sup> In silico simulation provides a unique avenue for generating spatially resolved ground-truth datasets with fully known parameters, enabling systematic evaluation of algorithmic performance, robustness, and scalability.<sup>19,20</sup> Existing ST simulators remain limited in scope. Reference-based models such as SRTsim preserve spatial expression patterns from specific datasets but fail to generalize across tissues with distinct geometries or cellular compositions.<sup>21–24</sup> Multimodal simulators like scDesign3 capture inter-modality correlations but are computationally intensive and difficult to scale to large modern ST datasets.<sup>25</sup> Deep-learning simulators such as scCube leverage extensive training data yet lack interpretability and fine control over biological perturbations.<sup>22</sup> De novo approaches such as scMultiSim can generate synthetic tissues from regulatory priors but remain challenging to calibrate for routine benchmarking.<sup>26</sup>

Additionally, a further gap lies in dimensionality: most simulators generate only two-dimensional (2D) slices, whereas current research increasingly demands quantitative evaluation of 3D reconstruction and spatiotemporal alignment methods.<sup>16,27</sup> Without realistic volumetric ground truths, assessment of 3D alignment and interpolation algorithms remains ad hoc and dataset-specific. This is more like a data augmentation problem: so how to use the most of these limited slices, to reconstruct realistic and biologically meaningful 3D region becomes to be a critical computational issue.

To overcome these challenges, we introduce FEAST (FEAture-space based modeling for Spatial Transcriptomics), a computational framework that models spatial transcriptomics data within a parameter cloud - a latent manifold capturing gene-level mean, variance, and sparsity. By sampling and perturbing this manifold, FEAST generates high-fidelity synthetic slices with tunable transcriptional and spatial alterations, enabling quantitative benchmarking of diverse analytical tasks including clustering, deconvolution, and spatial alignment. Beyond 2D simulation, FEAST performs 3D parameter-cloud interpolation across adjacent slices, integrating optimal transport and alignment-guided coordinate mapping to reconstruct missing sections and continuous 3D architectures for various downstream analyses. This process also supports controlled geometric perturbation and benchmark alignment, allowing quantitative evaluation of registration and reconstruction algorithms under known transformations.

Currently, FEAST is focusing on establishing a flexible and biologically interpretable simulation for comprehensive benchmarking and augmentation of spatial transcriptomics analyses. We anticipate that FEAST will serve as a reference framework for standardized 3D benchmarking simulation and data augmentation, providing a foundation for the next generation of computational spatial transcriptomics, and may be an extendable model for other tasks.

## 2 Methods

### 2.1 Single-slice simulation

To generate realistic ST data that preserve the distributional characteristics observed in real slices, FEAST builds a hierarchical generative framework operating at the gene level (Fig. 1). Specifically, each gene  $g$  is represented by a parameter vector  $\mathbf{p}_g = (\mu_g, \sigma_g^2, \pi_g)$ , capturing its mean, variance, and sparsity (zero proportion) within a single ST slice. These parameters are learned from empirical data and then used to simulate new, biologically plausible expression profiles.

**Concepts of Parameter cloud and feature space.** We first summarize the observed expression data into a compact **parameter cloud** that captures global gene-level behavior. The three-dimensional rectangular

coordinate system containing these points of the parameter cloud is called **feature space**. Given the gene expression count matrix  $\mathbf{X} \in \mathbb{R}_+^{N \times G}$ , where  $N$  denotes the number of cells (or spots) and  $G$  denotes the number of genes, we compute for each gene  $g$ :

$$\mu_g = \frac{1}{N} \sum_{i=1}^N X_{ig}, \quad \sigma_g^2 = \frac{1}{N} \sum_{i=1}^N (X_{ig} - \mu_g)^2, \quad \pi_g = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(X_{ig} = 0). \quad (1)$$

Here,  $\mu_g$  and  $\sigma_g^2$  represent the mean and variance of gene  $g$  across all spots, and  $\pi_g$  denotes the proportion of zero counts (dropout rate). The resulting cloud  $\{\mathbf{p}_g\}_{g=1}^G$  forms an empirical joint distribution over  $(\mu, \sigma^2, \pi)$ , summarizing how genes vary in expression level, variability, and sparsity within a single slice. This parameter cloud serves as the foundation for our generative model.

**Marginal models.** The next step models each component of the parameter cloud separately to capture the heterogeneity of gene-level statistics across the transcriptome. Each dimension (mean, variance, and sparsity) exhibits distinct statistical behaviors: expression means and variances often show heavy-tailed distributions, while sparsity proportions are bounded within  $[0, 1]$  and can be highly skewed. To flexibly represent these diverse patterns, we use mixture models to model the marginal distributions  $(f_\mu, f_{\sigma^2}, f_\pi)$ , which describe the individual variability of each parameter prior to dependency modeling. Detailed formulations are provided in Extended Methods S1.1.

**Joint dependency via C-vine copula.** To model nonlinear and asymmetric dependencies among  $(\mu, \sigma^2, \pi)$ , we employ a C-vine copula structure. Copulas allow us to model complex correlations among variables while preserving their fitted marginals. Let  $(F_\mu, F_{\sigma^2}, F_\pi)$  denote the marginal cumulative distribution functions, and define the corresponding uniform variables  $U_1 = F_\mu(\mu)$ ,  $U_2 = F_{\sigma^2}(\sigma^2)$ , and  $U_3 = F_\pi(\pi)$ . Under the C-vine construction, the joint density factorizes as (See Fig. 1a):

$$f(\mu, \sigma^2, \pi) = f_\mu(\mu) f_{\sigma^2}(\sigma^2) f_\pi(\pi) c_{12}(U_1, U_2) c_{13}(U_1, U_3) c_{23|1}(U_2|1, U_3|1), \quad (2)$$

where  $c_{ij}$  denotes the bivariate copula density between variables  $i$  and  $j$ , and  $U_{2|1} = \partial C_{12}(U_1, U_2) / \partial U_1$ ,  $U_{3|1} = \partial C_{13}(U_1, U_3) / \partial U_1$  are the conditional uniform variables. Each detailed pair-copula is selected based on the Bayesian Information Criterion (BIC) to balance model fit and parsimony. See details in Extended Methods S1.2.

This copula layer connects the three marginal models into a unified joint distribution  $f(\mu, \sigma^2, \pi)$  that preserves both the individual variability of each parameter and their interdependence. The resulting model thus represents a generative parameter space for one ST slice, from which new parameter triplets  $(\tilde{\mu}_g, \tilde{\sigma}_g^2, \tilde{\pi}_g)$  can be sampled. These sampled parameters are subsequently used to simulate gene-level expression profiles that mimic the empirical characteristics of the original data.

**Synthetic parameters and assignment.** From the fitted joint distribution obtained via the C-vine copula, we draw  $\alpha G$  synthetic parameter triplets  $\tilde{\mathbf{p}}^{(j)} = (\tilde{\mu}^{(j)}, \tilde{\sigma}^{2(j)}, \tilde{\pi}^{(j)})$ , where  $\alpha > 1$  controls the number of generated samples relative to the number of real genes  $G$ . To ensure biologically realistic correspondence, each synthetic triplet is assigned to a real gene through minimum-cost bipartite matching based on standardized features. The optimal one-to-one mapping  $\pi^*$  is obtained by solving the Hungarian algorithm. Further methodological details are provided in Extended Methods S1.3.

**Counts generation and assignment.** Having obtained the synthetic parameter triplets  $(\tilde{\mu}_g, \tilde{\sigma}_g^2, \tilde{\pi}_g)$  for each gene after assignment via Hungarian algorithm (see Details in Extended Methods S1.3), FEAST then generates spatially distributed expression counts that reflect the gene-specific variability. For each gene, an appropriate probabilistic model is selected from the Poisson, Negative Binomial (NB), Zero-Inflated Poisson (ZIP), or Zero-Inflated Negative Binomial (ZINB) families<sup>21, 28</sup> based on moments method and data-driven heuristics (See details in Extended Methods S1.4).

For Poisson and Negative Binomial distributions, the estimated parameters will be directly used into model. For ZIP( $\lambda, \pi$ ) and ZINB( $\mu, \alpha, \pi$ ) distributions, the parameters are estimated by numerically solving a system of equations that align the theoretical moments with the target moments derived from  $(\tilde{\mu}_g, \tilde{\sigma}_g^2, \tilde{\pi}_g)$ .

This moment-matching procedure ensures consistency between the simulated and empirical distributions (see Extended Methods S1.4 for implementation details). The theoretical moments are given by:

$$\text{ZIP: } E[X] = (1 - \pi)\lambda, \quad \text{Var}(X) = (1 - \pi)(\lambda + \pi\lambda^2) \quad (3)$$

$$\text{ZINB: } E[X] = (1 - \pi)\mu, \quad \text{Var}(X) = (1 - \pi)(\mu + \alpha\mu^2 + \pi\mu^2) \quad (4)$$

Once each gene's count distribution is parameterized, FEAST generates synthetic expression counts by random sampling from the selected model and spatially places them via rank-based assignment that preserves local spatial ranks and co-expression module structure (See details in Extended Methods S1.5).

## 2.2 Gene expression and spatial alteration

Let  $\mathbf{p}_g = (\mu_g, \sigma_g^2, \pi_g)$  and  $U_\mu = F_\mu(\mu_g)$ ,  $U_{\sigma^2} = F_{\sigma^2}(\sigma_g^2)$ ,  $U_\pi = F_\pi(\pi_g)$ . Given altered marginal distributions  $(F'_\mu, F'_{\sigma^2}, F'_\pi)$ , a copula-preserving transformation can be established as follows:

$$\mu'_g = (F'_\mu)^{-1}(U_\mu), \quad \sigma'^2_g = (F'_{\sigma^2})^{-1}(U_{\sigma^2}), \quad \pi'_g = (F'_\pi)^{-1}(U_\pi). \quad (5)$$

Detailed procedures for expression alteration are provided in Extended Methods S2. For spatial alteration, such as resolution adjustments and positional changes relevant to deconvolution and alignment tasks, please refer to Extended Methods S3-S4 and Fig. 1b.

## 2.3 3D slice interpolation

To reconstruct continuous 3D tissue structures from discrete ST slices, FEAST performs interpolation in the structured feature space rather than directly on noisy expression matrices. This strategy ensures that the interpolated slices preserve realistic gene-level statistical properties, including mean, variance, and sparsity, while maintaining smooth transitions of cellular and molecular patterns across the tissue depth.

The interpolation procedure combines optimal-transport (OT)-based parameter-cloud interpolation with transport-guided spatial coordinates and reference-guided expression reconstruction on the interpolated geometry. By integrating these three components, FEAST generates intermediate slices that are biologically coherent and spatially aligned with adjacent measured sections.

**Interpolation in feature space.** For two adjacent slices with parameter clouds  $\mathcal{C}_k$  and  $\mathcal{C}_{k+1}$ , we define empirical measures over the gene-level parameter triplets  $\mathbf{p}_{g,k} = (\mu_{g,k}, \sigma_{g,k}^2, \pi_{g,k})$ .

The parameter cloud  $\mathcal{C}_k = \{\mathbf{p}_{i,k}\}_{i=1}^G$  can be viewed as a discrete empirical measure  $\nu_k$  in  $\mathbb{R}^3$ , where each point corresponds to one gene's statistical signature. For the adjacent slice  $k + 1$ , the corresponding cloud  $\mathcal{C}_{k+1} = \{\mathbf{p}_{j,k+1}\}_{j=1}^G$  defines another empirical measure  $\nu_{k+1}$ . Together, these clouds represent the global gene-level landscapes of two consecutive slices, providing a high-level view of molecular variation across tissue depth.

To generate an intermediate slice positioned fractionally between the two, we compute the 2-Wasserstein barycenter of  $\nu_k$  and  $\nu_{k+1}$  for an interpolation coefficient  $t \in (0, 1)$ :

$$\nu_t = \arg \min_{\nu} (1 - t)W_2^2(\nu_k, \nu) + tW_2^2(\nu_{k+1}, \nu), \quad (6)$$

where  $W_2^2$  denotes the squared 2-Wasserstein distance between probability measures. Intuitively,  $\nu_t$  represents the optimal intermediate distribution along the Wasserstein geodesic between slices  $k$  and  $k + 1$ , ensuring a smooth and mass-preserving transition between their parameter distributions.

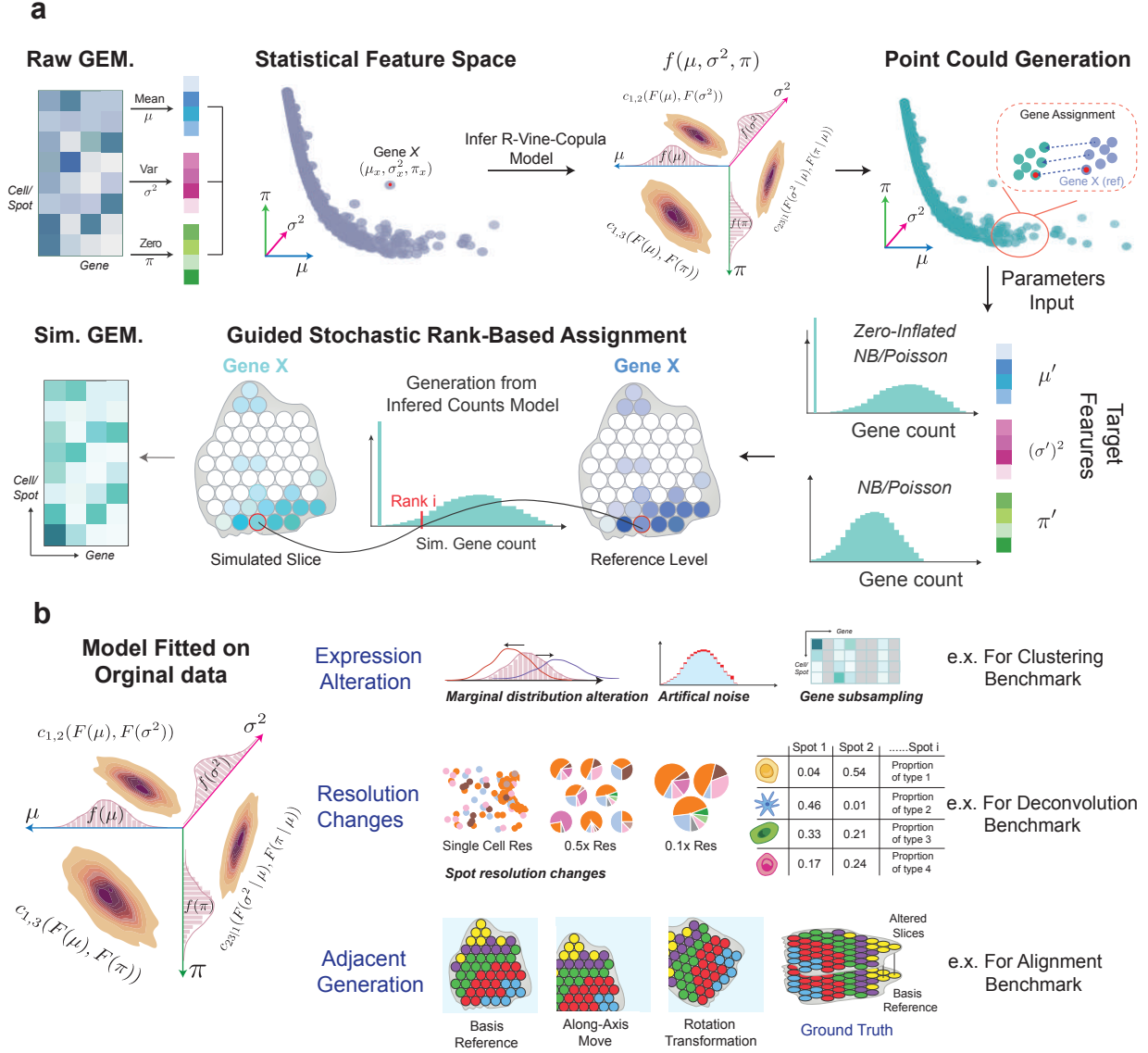
The barycenter  $\nu_t$  is realized by displacement interpolation under the OT plan  $\mathbf{T}^*$  with quadratic cost, producing interpolated parameters for each gene index  $i$  in slice  $k$ :

$$\mathbf{p}_{i,t} = (1 - t)\mathbf{p}_{i,k} + t \sum_j \frac{T_{ij}^*}{\sum_\ell T_{i\ell}^*} \mathbf{p}_{j,k+1}, \quad (7)$$

here,  $\mathbf{T}^*$  is obtained once by minimizing the total quadratic transport cost between the two clouds:

$$\mathbf{T}^* = \arg \min_{\mathbf{T} \geq 0} \sum_{i,j} T_{ij} \|\mathbf{p}_{i,k} - \mathbf{p}_{j,k+1}\|_2^2. \quad (8)$$





**Fig. 1. Overview of the FEAST framework.** **a**, Core single-slice simulation pipeline. FEAST learns a parameter cloud describing gene-wise mean, variance, and sparsity from a raw gene expression matrix (GEM). A C-vine copula captures the dependency structure among these parameters, enabling the generation of new parameter sets that preserve their joint relationships. These simulated parameters are assigned to genes under a chosen count model to produce synthetic expression profiles, followed by a rank-based mapping to construct the simulated GEM. **b**, Modules for generating benchmark datasets through controlled alteration of expression distributions, spatial resolutions, or spatial transformations to create adjacent slices.

subject to marginal constraints  $\sum_j T_{ij} = a_i$  and  $\sum_i T_{ij} = b_j$ .  $T_{ij}^*$  denotes the optimal transport weight between parameter point  $i$  in slice  $k$  and point  $j$  in slice  $k+1$ , defining how much “mass” of gene  $i$  is moved toward gene  $j$ . The denominator  $\sum_\ell T_{i\ell}^*$  normalizes the mapping for each  $i$ . In this construction, the index  $i$  refers to a gene (or feature) in the source slice,  $j$  indexes its potential matches in the target slice, and  $t$  controls the interpolation position between the two. The interpolation coefficient  $t$  does not affect this optimization; instead,  $t$  determines how far each source point moves along its transport path after  $\mathbf{T}^*$  is fixed. The resulting  $\mathbf{p}_{i,t}$  therefore describes a continuous trajectory between  $\mathbf{p}_{i,k}$  and its matched destinations in slice  $k+1$ .

The resulting interpolated cloud  $\mathcal{C}_t = \{\mathbf{p}_{i,t}\}$  defines the synthetic parameter distribution of an intermediate slice. This parameter cloud serves as the generative foundation for producing virtual ST slices using the single-slice simulation framework, enabling dense 3D continuity for downstream alignment, clustering, and full tissue reconstruction.

**Alignment-guided coordinates.** Following parameter-cloud interpolation, the next step focuses on spatial geometry, interpolating the coordinates of tissue spots between consecutive slices in a way that is consistent with their molecular alignment. We begin with a pre-computed alignment matrix generated by Spateo<sup>16</sup>  $\mathbf{T}_{\text{align}} \in \mathbb{R}_+^{N_k \times N_{k+1}}$ , which represents the probabilistic transport plan between spatial spots of adjacent slices  $k$  and  $k+1$ .

Each element  $T_{\text{align},ij}$  quantifies the alignment confidence between spot  $i$  in slice  $k$  and spot  $j$  in slice  $k+1$ . From this matrix, we extract a high-confidence set of correspondences to guide the interpolation of spatial coordinates. This ensures the interpolated geometry preserves biologically meaningful correspondences implied by the molecular transport.

For each high-confidence correspondence, we then generate interpolated coordinates using linear displacement interpolation between the two matched spots:

$$\mathbf{s}_{i,t} = (1-t)\mathbf{s}_{i,k} + t\mathbf{s}_{j,k+1}. \quad (9)$$

Here,  $\mathbf{s}_{i,k}$  and  $\mathbf{s}_{j,k+1}$  are the 2D spatial coordinates of matched spots in slices  $k$  and  $k+1$ , respectively, while  $t \in [0, 1]$  determines the relative position of the interpolated point along the displacement path between them. When  $t = 0$ , the coordinates correspond exactly to slice  $k$ ; when  $t = 1$ , they match slice  $k+1$ ; and for  $0 < t < 1$ , they yield intermediate spatial positions that ensure smooth geometric continuity across slices.

This interpolation process can be viewed as a spatial analog to the parameter-cloud interpolation: rather than transporting gene-level parameters, it smoothly transports the spatial coordinates of corresponding tissue spots.

**Ordered query slice.** After the gene-level parameters are interpolated through Eq. (7) and the single-slice simulation framework generates corresponding gene count matrices, FEAST can assign these counts onto the interpolated geometry using the previously mentioned rank-based spatial assignment procedure. Because this placement requires reference information from both adjacent slices  $k$  and  $k+1$ , FEAST first constructs a unified query slice that aggregates transcriptomic and spatial features from both slices. This query slice serves as a reference layer, encoding expected gene expression patterns across the interpolated coordinates and ensuring that spatial transitions between slices remain smooth and biologically coherent.

To build the gene expression profile for this query slice, FEAST initializes expression at each interpolated spot by blending the aligned source profiles in normalized or log-transformed space. Let  $w = T_{\text{align},ij}$  denote the alignment weight between spot  $i$  in slice  $k$  and spot  $j$  in slice  $k+1$ . The weighted blending is defined as:

$$\tilde{w}_1 = (1-t)w, \quad \tilde{w}_2 = tw, \quad \omega_1 = \frac{\tilde{w}_1}{\tilde{w}_1 + \tilde{w}_2}, \quad \omega_2 = \frac{\tilde{w}_2}{\tilde{w}_1 + \tilde{w}_2}, \quad (10)$$

$$\mathbf{X}_{i,t}^{\text{query}} = \omega_1 \mathbf{X}_{i,k}^{(\text{norm})} + \omega_2 \mathbf{X}_{j,k+1}^{(\text{norm})}. \quad (11)$$

Here,  $\mathbf{X}_{i,k}^{(\text{norm})}$  and  $\mathbf{X}_{j,k+1}^{(\text{norm})}$  denote the normalized gene expression vectors of matched spots from slices  $k$  and  $k+1$ . The coefficients  $(\omega_1, \omega_2)$  define a convex interpolation between these two expression profiles according to the interpolation depth  $t$ . Because the alignment weight  $w$  cancels in Eq. (10), the resulting blend depends solely on  $t$ , producing a smooth transition of expression intensity across depth.

**Finalization.** FEAST then uses this query slice with  $\mathbf{X}_{i,t}^{\text{query}}$  as the spatial-molecular reference for finalizing rank-based gene count assignment, ensuring that each simulated count is positioned according to its anatomically coherent location and biologically consistent expression context.

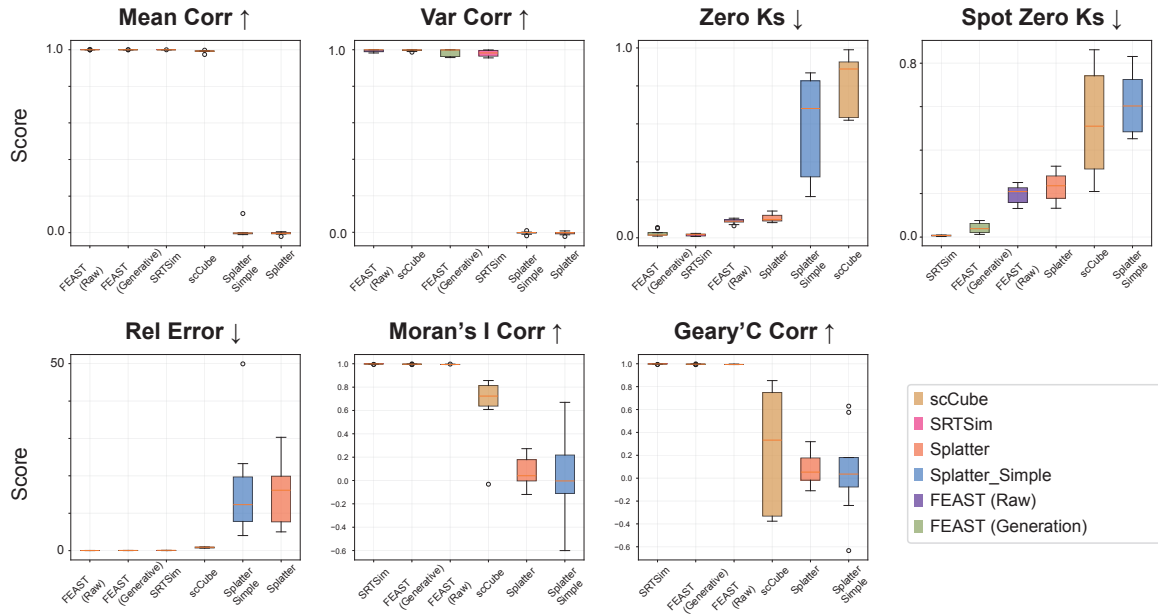
## 3 Results

### 3.1 FEAST generates high-fidelity spatial transcriptomics slices

To comprehensively assess the performance of FEAST, we benchmarked it against several state-of-the-art spatial and single-cell simulators, including SRTsim, scCube, and the Splatter series. We generated syn-

thetic datasets spanning a broad spectrum of ST platforms and biological contexts, including 10x Visium,<sup>29</sup> MERFISH,<sup>30</sup> OpenST,<sup>31</sup> Slide-seqV2, Xenium, and Stereo-seq, and systematically evaluated the fidelity of FEAST outputs in capturing gene-level expression statistics, spot-level properties, and spatial expression patterns (Fig. 1; See Extended Methods S5.1).

To isolate the contribution of FEAST’s parameter cloud independent of its generative process, we implemented a streamlined mode that directly employs the extracted parameters without resampling, referred to as FEAST (Raw). The full generative framework is denoted as FEAST (Generative). Benchmark analyses revealed that two modes achieve superior or highly competitive performance across all evaluation criteria (Fig. 2). In preserving fundamental gene-wise statistics, FEAST achieved near-perfect correlation for both gene means and variance estimates. Furthermore, it accurately modeled the characteristic sparsity of ST data, as reflected by minimal Kolmogorov-Smirnov (KS) distances in gene-level zero proportions and spot-level zero proportions. Beyond expression fidelity, FEAST effectively recapitulated complex spatial architectures, achieving Moran’s I and Geary’s C correlations nearly identical to those of the ground-truth reference datasets, thereby demonstrating its ability to preserve both global and local spatial dependencies intrinsic to tissue organization. Our visualization of marker gene expression between experimental and FEAST-generated slices across multiple ST platforms revealed strong concordance in spatial localization and expression intensity patterns (Supplementary Fig. 1). It is worth noting that while SRTsim achieved overall performance comparable to FEAST, this outcome is largely expected because, in its reference-based simulation mode, SRTsim directly derives gene-wise and spatial distribution parameters from the real reference data. In contrast, FEAST re-learns these parameters de novo through its parameter cloud representation. Consequently, FEAST attains similar or superior fidelity while offering substantially greater flexibility and generalizability across spatial platforms and tissue architectures.



**Fig. 2. Quantitative benchmarking of FEAST against existing simulators.** Boxplots summarize performance across multiple datasets using seven key evaluation metrics: gene-wise mean correlation, variance correlation, zero-proportion Kolmogorov-Smirnov (KS) distance, spot-level zero KS distance, relative count error, and spatial autocorrelation (Moran’s I and Geary’s C). Arrows indicate the direction of improvement for each metric (↑ = higher is better, ↓ = lower is better).

### 3.2 Benchmarking downstream single-slice analysis tools

A central motivation for developing simulation frameworks is to enable systematic and interpretable benchmarking of downstream spatial analysis algorithms. Leveraging FEAST’s controllable data alteration capabilities (Fig. 1b), we generated benchmark datasets with tunable ground-truth perturbations to evaluate clustering and deconvolution performance on single-slice ST data.

**Robustness analysis of spatial clustering methods.** We assessed the robustness of two state-of-the-art spatial clustering algorithms, STAGATE<sup>8</sup> and GraphST,<sup>10</sup> by systematically perturbing specific statistical properties of a simulated dataset with known ground truth. Using a 10x Visium human DLPFC slice as the reference, FEAST synthesized altered datasets by independently modifying gene-wise mean expression and sparsity levels while maintaining the underlying joint dependency structure across genes (see Extended Methods S2). The alteration factor of 1.0 corresponds to the unaltered (reference) condition, whereas values below or above 1.0 represent controlled perturbations that are expected to reduce performance as deviation from the original distribution increases. Across the mean-alteration series, STAGATE exhibited higher clustering accuracy (ARI and NMI) and stronger spatial continuity (PAS and CHAOS), maintaining stable performance across a wide dynamic range of mean-expression perturbations. GraphST, in contrast, showed faster degradation as the alteration factor deviated from 1.0. Under sparsity perturbations, STAGATE again demonstrated superior robustness, preserving coherent spatial domains even under high-dropout conditions (Fig. 3). Collectively, these controlled perturbation experiments underscore FEAST’s utility as a principled benchmarking framework: by selectively altering key statistical factors - mean, variance, and sparsity - while preserving spatial organization, FEAST enables mechanistic dissection of model robustness and delineates the operational boundaries of current spatial clustering algorithms.

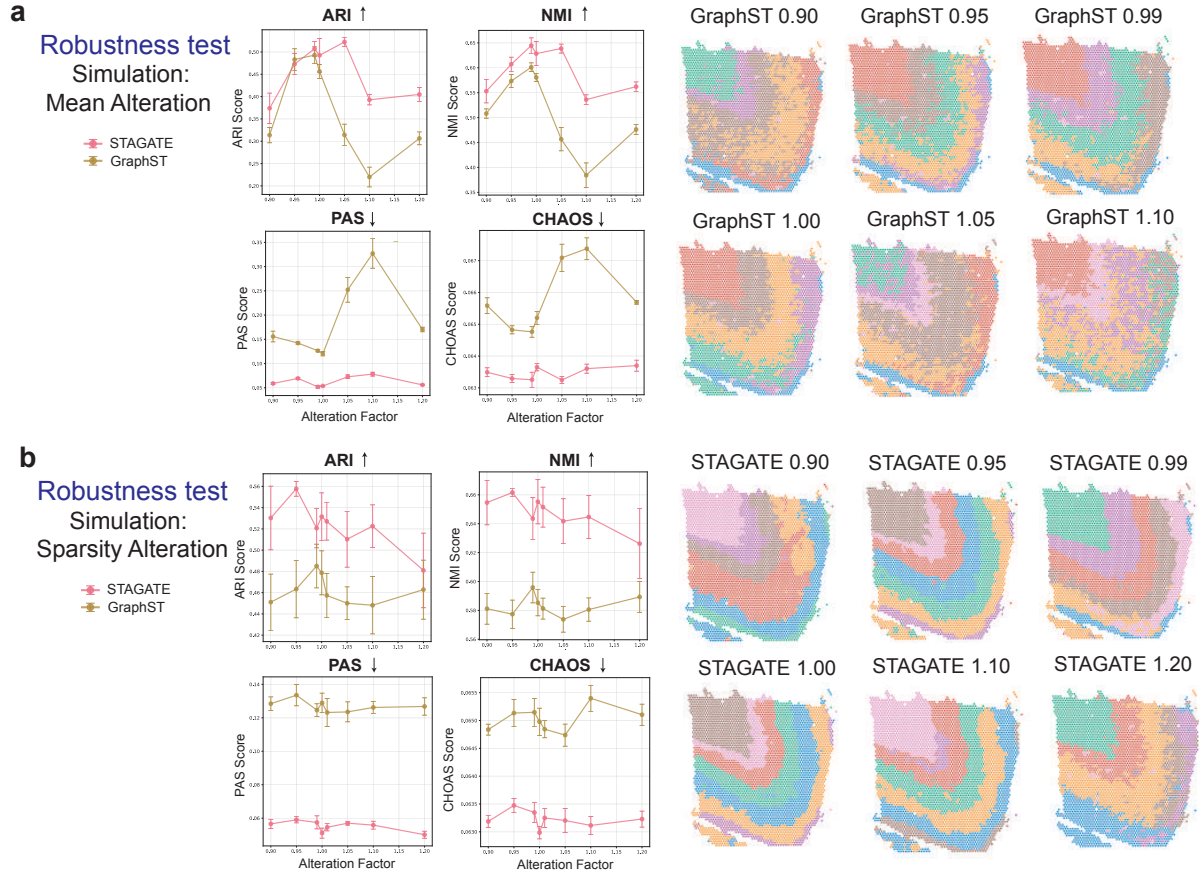
**Generating ground-truth data for cell-type deconvolution.** Robust benchmarking of spatial deconvolution methods requires datasets in which the true cell-type composition of each spot is known, a resource that is rarely available for real ST experiments and is typically approximated using non-spatial pseudo-bulk mixtures. FEAST fills this gap by deriving realistic multi-resolution spot-based datasets from single-cell-resolution spatial maps while retaining full knowledge of the underlying cellular mixtures. Starting from a high-resolution MERFISH slice,<sup>30</sup> we use FEAST to aggregate single cells into larger synthetic spots on user-defined grids to generate corresponding ST slices at coarser resolutions, recording the exact cell-type proportions per spot as ground truth (Supplementary Fig. 2). In addition, FEAST can introduce controlled gene-level perturbations (e.g., mean and sparsity alterations as in our clustering benchmarks), enabling systematic evaluation of the robustness of current spatial deconvolution tools, such as Cell2location,<sup>32</sup> across both resolution changes and expression regimes.

### 3.3 Controlled benchmarks for spatial alignment algorithms

Integrating serial tissue sections requires accurate alignment across slices, yet quantitative benchmarking of alignment algorithms remains limited due to the lack of datasets with known spatial correspondences. Most existing evaluations rely on heuristic visual inspection or region-level concordance rather than true spot-to-spot accuracy. FEAST overcomes this limitation by generating paired datasets in which one slice is a precisely controlled geometric transformation, such as rotation, translation, or elastic deformation, of a reference slice while preserving its molecular and spatial context (Extended Methods S3).

To illustrate, we simulated a series of controlled rotations (1°, 5°, 10°, 30°, 45°, and 60°) on a 10x Visium DLPFC slice and systematically compared two state-of-the-art alignment algorithms, Spateo<sup>16</sup> and SPACEL.<sup>33</sup> Across multiple evaluation metrics, including Accuracy, Precision, F1 Score, Gene Expression Correlation, and Adjusted Region Accuracy, Spateo consistently outperformed SPACEL (Fig. 4). Spateo maintained near-perfect alignment accuracy and high gene-level correspondence up to a 30° rotation, whereas SPACEL exhibited a rapid decline in performance under moderate to severe transformations (Supplementary Fig. 3).

These results demonstrate how FEAST enables quantitative and reproducible assessment of spatial alignment methods under well-defined perturbations, offering a benchmark framework that exposes algorithmic robustness, sensitivity to geometric distortion, and performance boundaries across diverse spatial contexts.

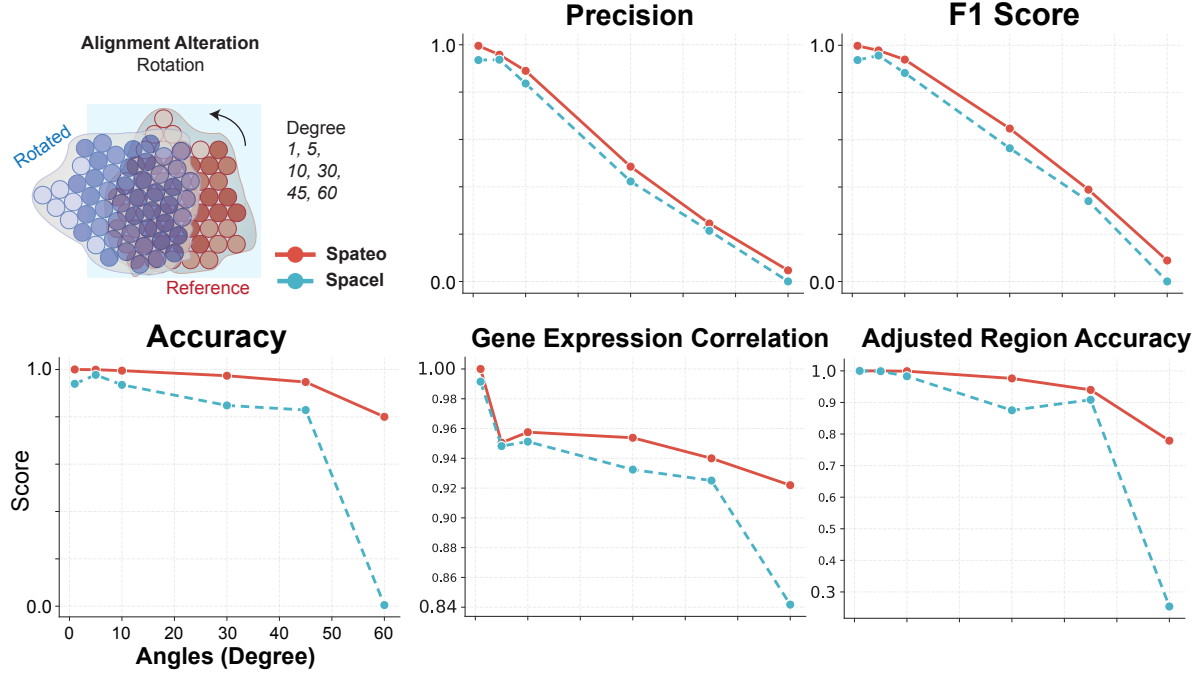


**Fig. 3. Benchmarking spatial clustering algorithms using controlled expression and sparsity perturbations.** **a**, Robustness testing under mean alteration. FEAST generated simulated datasets from a 10x Visium DLPFC slice by systematically modifying gene-wise mean expression, where alteration factors below or above 1.0 represent down- or up-scaling. Line plots show clustering performance of STAGATE (pink) and GraphST (gold) evaluated by Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and spatial continuity metrics (PAS, CHAOS). Spatial maps visualize clustering outcomes under increasing perturbation magnitudes. The alteration factor of 1.0 (denoted by \*) represents the unaltered reference condition. **b**, Robustness testing under sparsity alteration. FEAST introduced controlled changes in dropout rate to assess model stability.

### 3.4 3D interpolation recovers missing slices with high fidelity

Reconstructing continuous three-dimensional tissue volumes from sparsely profiled ST slices remains a key challenge in spatial biology, as most experimental platforms capture only a limited number of serial sections. To address this, FEAST provides a quantitative framework for evaluating interpolation performance under realistic conditions using ground-truth experimental data (Fig. 5a). We assessed its 3D interpolation capability on MERFISH datasets<sup>30</sup> through a leave-one-out validation design, where middle slices were sequentially withheld and reconstructed solely from their neighboring experimental slices. This setup enables direct and quantitative comparison between interpolated and experimentally measured slices.

Using five consecutive MERFISH sections (slice IDs 5-9), FEAST reconstructed the omitted middle slices (IDs 6-8) by first aligning adjacent experimental slices, computing optimal transport-based correspondences between parameter clouds of gene-level statistics, and performing smooth interpolation in both spatial and expression domains to regenerate the missing slices. The reconstructed slices recapitulated key anatomical and molecular features, with marker genes showing highly concordant spatial distributions relative to experimental ground truth (Fig. 5b).



**Fig. 4. Benchmarking spatial alignment algorithms using controlled geometric transformations.** FEAST generated paired slices in which one slice was rotated relative to the reference by defined angles ( $1^\circ$ ,  $5^\circ$ ,  $10^\circ$ ,  $30^\circ$ ,  $45^\circ$ ,  $60^\circ$ ) to create datasets with known ground-truth correspondences. Line plots compare the alignment performance of Spateo (red) and SPACEL (blue) across five quantitative metrics: Precision, F1 Score, Accuracy, Gene Expression Correlation, and Adjusted Region Accuracy.

Quantitatively, the interpolated slices exhibited strong agreement with experimental data across multiple criteria: mean gene expression correlation and variance correlation both exceeded 0.9, Moran’s I correlation confirmed preservation of spatial autocorrelation structure, and Kolmogorov-Smirnov tests of zero proportion indicated consistent sparsity distributions (Fig. 5c). Additionally, perceptual hash similarity scores corroborated visual fidelity between reconstructed and real slices, validating both molecular and spatial coherence.

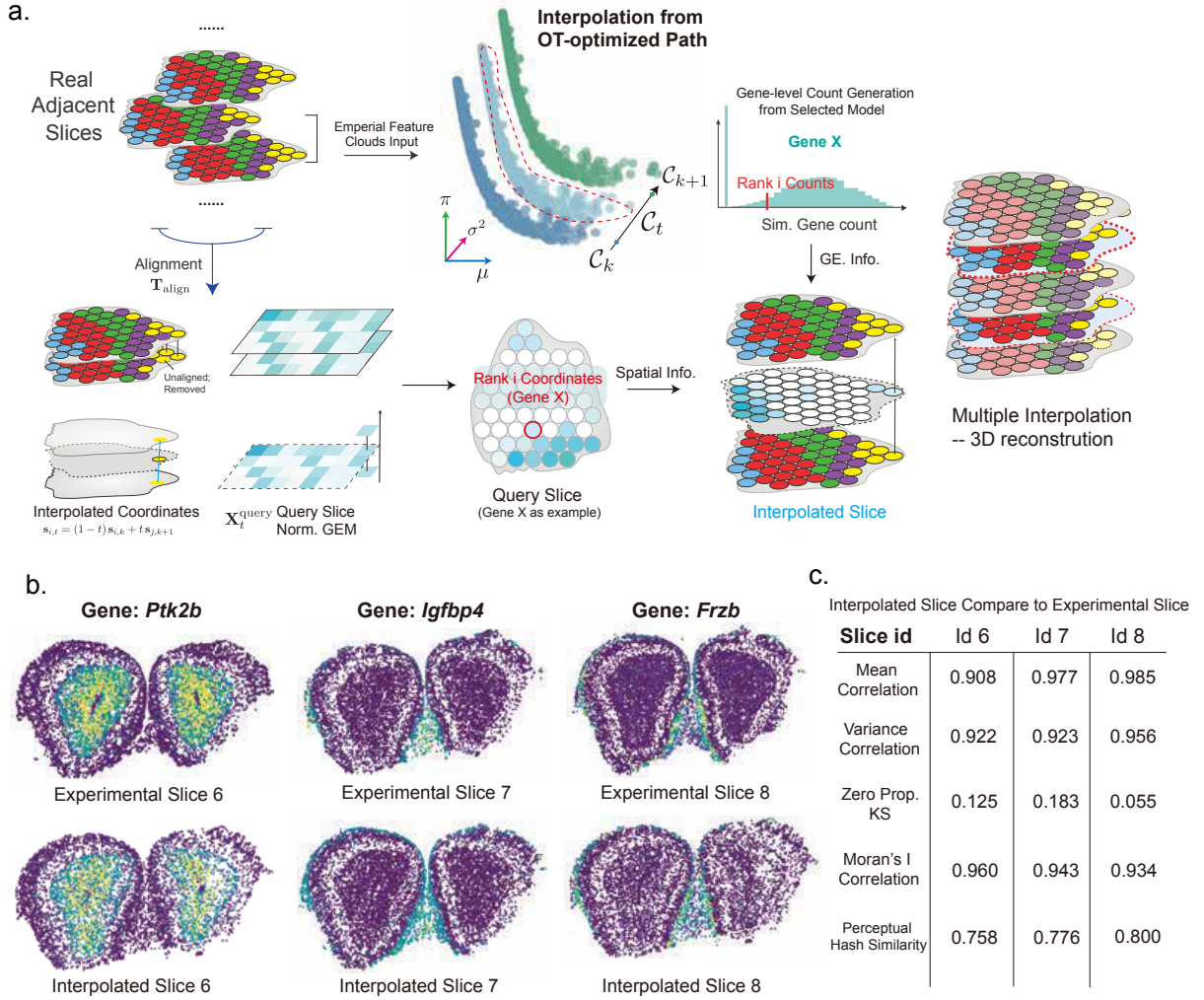
We further evaluated our interpolation framework by reconstructing full 3D tissue architectures from sparsely sampled slices. Using two MERFISH spinal cord sections spaced  $250\mu\text{m}$  apart as anchors, FEAST-sim generated nine intermediate slices, producing a coherent volumetric reconstruction with smoothly continuous gene-expression patterns (Supplementary Fig. 4). These results illustrate that our approach robustly preserves continuous gene-expression transitions and structural integrity across reconstructed 3D tissues.

Importantly, FEAST’s interpolation module is platform-agnostic and generalizes to any other spatial transcriptomic modalities. By operating on normalized expression matrices and spatial coordinates within a shared parameter-cloud representation, the framework enables unified evaluation and reconstruction across diverse experimental resolutions. Together, these results demonstrate that FEAST can faithfully recover missing spatial layers, facilitating high-fidelity 3D reconstruction of complex tissues from limited or partially sampled spatial data.

## 4 Discussion and Conclusion

FEAST provides a unified and interpretable modeling framework for simulating, benchmarking, and 3D interpolation. By modeling data within a parameter-cloud manifold that encodes gene-level mean, variance, and sparsity while preserving higher-order dependencies, FEAST bridges statistical realism with spatial structure. This framework supports a wide range of applications, including single-slice simulation, ground-truth gener-





**Fig. 5. Overview of the FEAST interpolation framework and benchmarking results.** **a**, Overview of the FEAST interpolation framework. Details are provided in the Methods section. **b**, Visualization of reconstructed versus experimental slices for representative genes (*Ptk2b*, *Igfbp4*, *Frzb*) from the MERFISH dataset. Interpolated slices closely recapitulate spatial gene expression patterns observed in experimental data. **c**, Quantitative comparison between interpolated and experimental slices (IDs 6-8), showing high concordance across mean and variance correlation, sparsity consistency (zero proportion KS), spatial autocorrelation (Moran's I), and perceptual hash similarity.

ation for clustering and deconvolution, alignment benchmarking under controlled geometric transformations, and 3D interpolation for volumetric reconstruction.

## Funding

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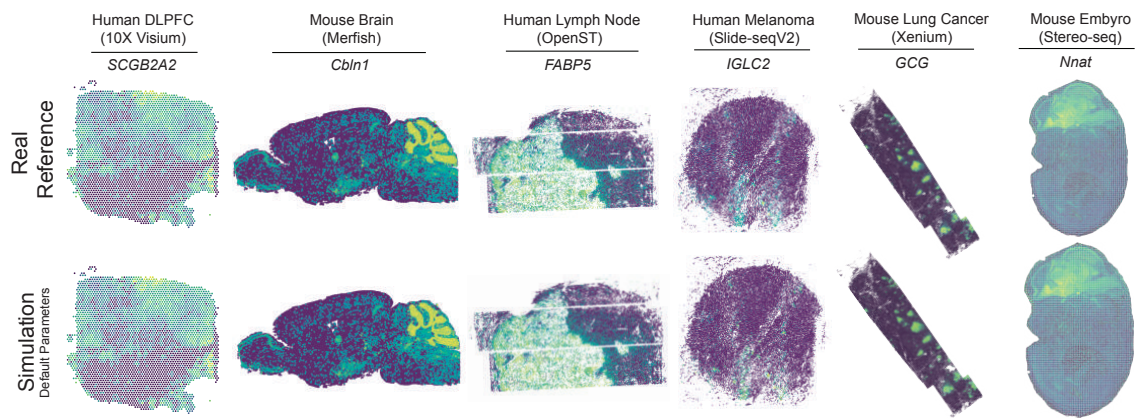
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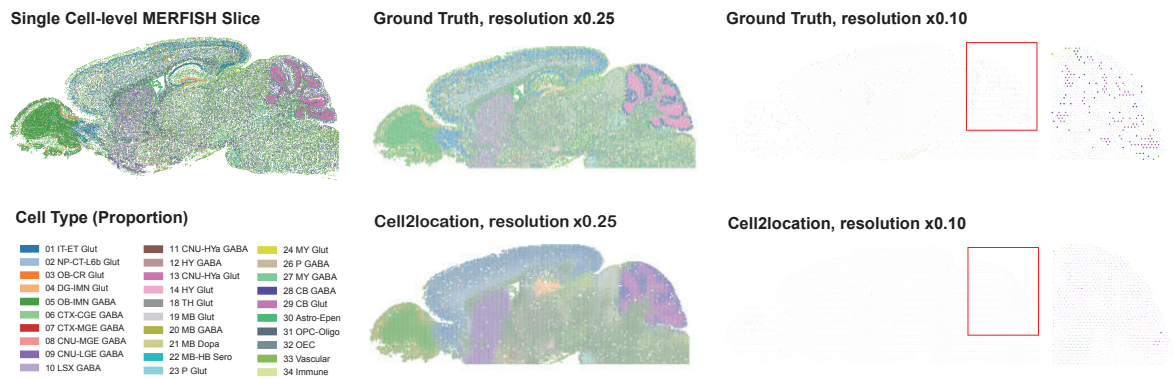


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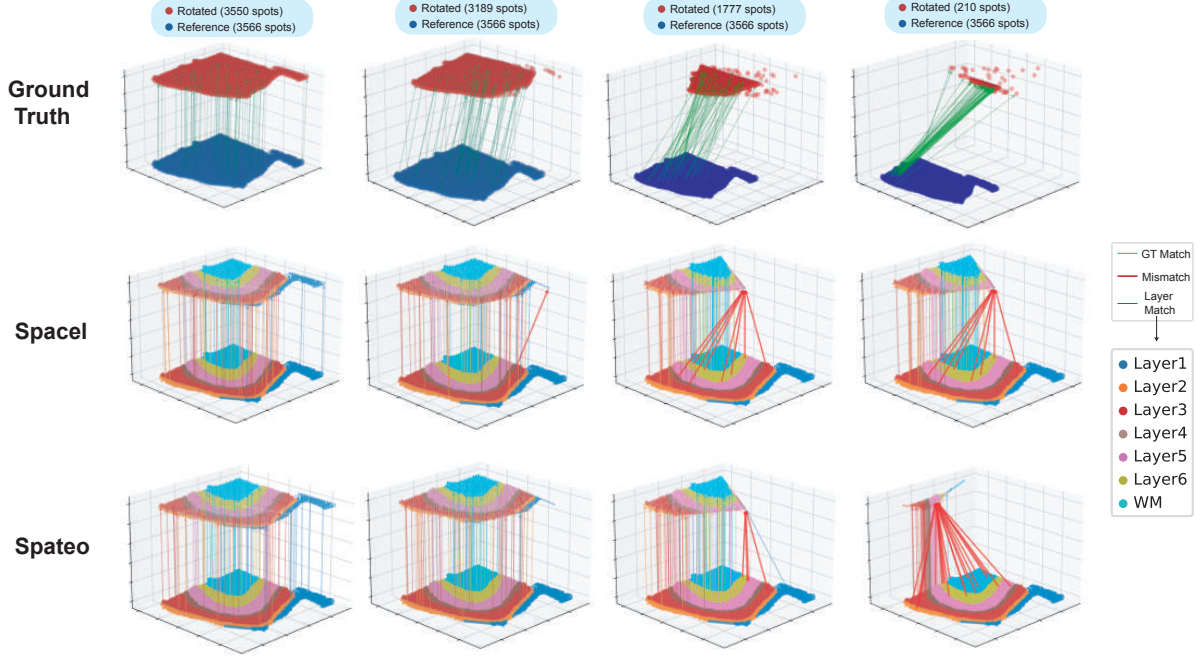
Supplementary Materials: Supplementary Figures



**Supplementary Fig. 1. Visualization of marker gene concordance between real and simulated slices across multiple spatial transcriptomic platforms.** Representative marker genes are shown for six datasets spanning different sequencing technologies: human DLPFC (10x Visium, SCGB2A2), mouse brain (MERFISH, Cbln1), human lymph node (OpenST, FABP5), human melanoma (Slide-seqV2, IGLC2), mouse lung cancer (Xenium, GCG), and mouse embryo (Stereo-seq, Nnat). Simulated slices generated by FEAST under default parameters recapitulate the spatial localization and expression intensity of marker genes observed in real reference datasets, demonstrating high cross-platform fidelity.



**Supplementary Fig. 2. Benchmarking cell-type deconvolution using simulated datasets across multiple spatial resolutions.** FEAST generates synthetic ST slices with multi-cellular spots derived from single-cell-resolution MERFISH data. It displays simulated ground-truth cell-type proportion maps at two coarser spatial resolutions (0.25 $\times$  and 0.10 $\times$  of the original) together with corresponding Cell2location deconvolution results. The simulated slices preserve realistic spatial organization and cell-type mixtures while providing quantitative ground truth for evaluating and calibrating cell-type deconvolution tools across spatial scales.



**Supplementary Fig. 3. Illustration of spatial alignment simulation and evaluation.** FEAST generated paired slices with controlled rotational transformations to create ground-truth correspondences for alignment benchmarking. The top row shows ground-truth mappings between reference (blue) and rotated (red) slices, where green lines indicate correct spot-to-spot matches. The middle and bottom rows display alignment results from SPACEL and Spateo, respectively. Correct matches inferred by each algorithm are shown in blue, while incorrect mappings are highlighted in red. Each layer (Layer 1-6 and white matter, WM) is color-coded according to anatomical position. As the rotation angle increases from left to right, mismatches accumulate, particularly in SPACEL, whereas Spateo maintains more accurate cross-layer alignment, demonstrating its robustness to geometric transformations.

## Supplementary Materials: Extended Methods

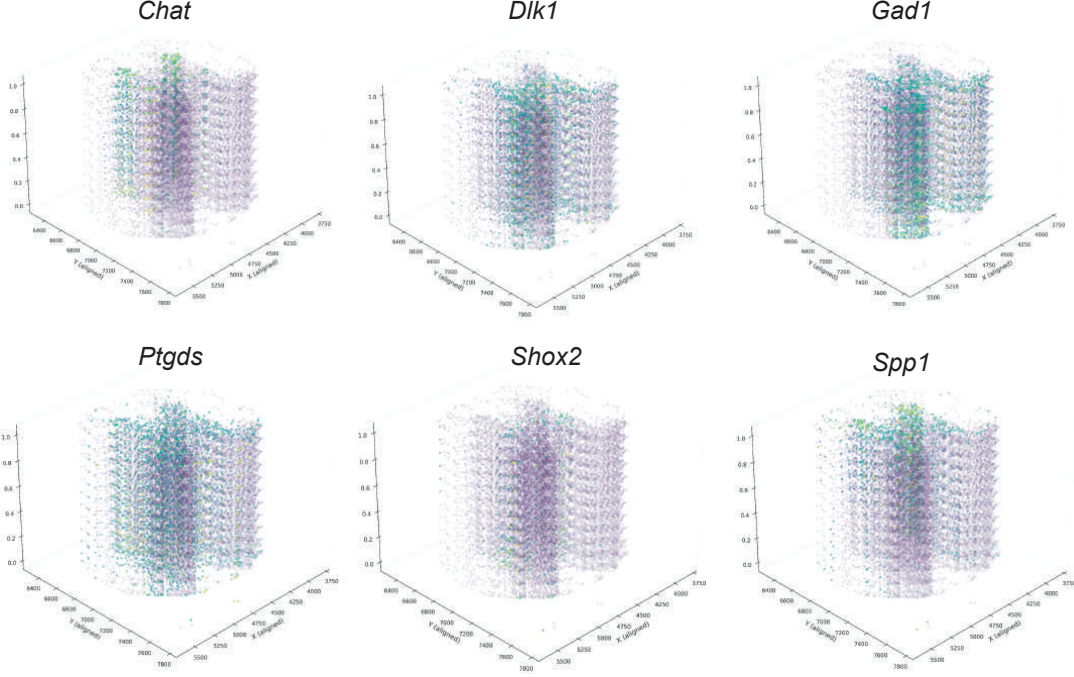
### S1 Single-slice Simulation Details

**S1.1 Marginal Model Formulations.** In the marginal mixture models, the mean, variance, and sparsity parameters of each gene are independently modeled to capture their empirical distributions across the dataset. Specifically, the mean and variance parameters are modeled using Student- $t$  mixtures. The density for the mean parameter is shown below; the density for the variance parameter has a similar form and is therefore omitted. The sparsity  $\pi \in [0, 1]$  is modeled using Beta mixtures. Each mixture contains up to 10-15 components by default to balance flexibility and computational efficiency:

$$f(\mu) = \sum_{k=1}^{K_\mu} w_k^{(\mu)} t(\mu; \nu_k, \theta_k, \sigma_k), \quad t(x; \nu, \theta, \sigma) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\nu\pi}\sigma} \left(1 + \frac{(x-\theta)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}, \quad (12)$$

$$f(\pi) = \sum_{k=1}^{K_\pi} w_k^{(\pi)} \text{Beta}(\pi; \alpha_k, \beta_k), \quad \text{Beta}(x; \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}. \quad (13)$$

In the above formulations,  $w_k^{(\mu)}$  and  $w_k^{(\pi)}$  denote the mixture weights satisfying  $\sum_k w_k^{(\cdot)} = 1$ . For the Student- $t$  components,  $\theta_k$  is the location parameter,  $\sigma_k$  the scale, and  $\nu_k$  the degrees of freedom controlling tail heaviness. This distribution is well-suited for modeling overdispersed and heavy-tailed behavior commonly observed in gene expression means and variances with decent complexity while remaining accuracy.



**Supplementary Fig. 4. 3D region reconstruction using two spinal cord MERFISH slices** FEAST generated nine intermediate slices from two reference sections to reconstruct the local 3D spinal cord region. Genes exhibiting similar patterns and expression levels across the two references show smoothly continuous spatial patterns in the interpolated slices. For genes with more variable expression, such as *Spp1*, the model learns and produces a biologically plausible smooth gradient across the reconstructed volume.

For Beta mixtures modeling sparsity: in the Beta mixture,  $\alpha_k$  and  $\beta_k$  are the shape parameters that determine the skewness and concentration of each component, allowing the model to represent multimodal or asymmetric sparsity patterns across genes. The Beta distribution naturally constrains the sparsity parameter  $\pi \in [0, 1]$ , making it ideally suited for modeling proportions.

The output of this stage is a set of fitted marginal distributions  $(f_\mu, f_{\sigma^2}, f_\pi)$ , which describe the individual variability of each parameter independently. However, these parameters are not statistically independent. Genes with high mean expression tend to have higher variance and lower sparsity. To capture such dependencies, a copula-based dependency model is subsequently introduced in the Methods section.

**S1.2 C-Vine Copula Selection Criteria.** Each pair-copula  $c_{ij}$  or  $c_{ij|k}$  can belong to a different family (e.g., Gaussian, Clayton), allowing flexible modeling of tail dependence and non-Gaussian relationships. The optimal copula family for each pair is selected based on the Bayesian Information Criterion (BIC) to balance model fit and parsimony. This flexibility is crucial because different variable pairs (mean-variance, mean-sparsity, variance-sparsity) may exhibit distinct dependency structures that are best captured by different copula families.

**S1.3 Synthetic Parameter Assignment via Bipartite Matching.** Let  $\bar{\mu}_g, \bar{\sigma}_g^2, \bar{\pi}_g$  denote the log-scaled,  $z$ -scored statistics for gene  $g$ , and  $(\bar{\mu}^{(j)}, \bar{\sigma}^{2(j)}, \bar{\pi}^{(j)})$  denote those for synthetic sample gene  $j$ . We define the cost matrix as:

$$C_{g,j} = \sqrt{w_\mu(\bar{\mu}_g - \bar{\mu}^{(j)})^2 + w_{\sigma^2}(\bar{\sigma}_g^2 - \bar{\sigma}^{2(j)})^2 + w_\pi(\bar{\pi}_g - \bar{\pi}^{(j)})^2}, \quad (14)$$

where  $(w_\mu, w_{\sigma^2}, w_\pi)$  are the feature weights that emphasize mean expression as the dominant matching criterion. By default, these weights are set to prioritize the mean parameter while allowing variance and

sparsity to contribute to the matching distance. This weighting scheme reflects the biological importance of the mean expression level in determining gene behavior.

The optimal one-to-one mapping  $\pi^*$  between real and synthetic genes is obtained by solving a global minimum-cost assignment problem using the Hungarian algorithm:

$$\pi^* = \arg \min_{\pi} \sum_{g=1}^G C_{g, \pi(g)}. \quad (15)$$

This assignment aligns each gene in the original slice with its most similar synthetic counterpart in the parameter cloud, ensuring that downstream simulations maintain realistic per-gene variability and dropout characteristics. The Hungarian algorithm guarantees a globally optimal matching in polynomial time, making this approach computationally tractable even for datasets with thousands of genes.

**S1.4 Count Generation: Theoretical Moments and Parameter Optimization.** For each gene, an appropriate probabilistic model is selected from the Poisson, Negative Binomial (NB), Zero-Inflated Poisson (ZIP), or Zero-Inflated Negative Binomial (ZINB) families based on data-driven heuristics of zero-inflation and overdispersion (default thresholds:  $\pi_g > 0.3$  for zero-inflation and  $(\sigma_g^2/\mu_g) > 1.5$  for overdispersion). This adaptive selection ensures that genes with sparse, overdispersed expression are modeled more flexibly, while highly expressed and stable genes are represented with simpler distributions. The selection logic is summarized as follows:

- If  $\pi_g \leq 0.3$  and  $\sigma_g^2/\mu_g \leq 1.5$ : use Poisson (equidispersed, no zero-inflation)
- If  $\pi_g \leq 0.3$  and  $\sigma_g^2/\mu_g > 1.5$ : use Negative Binomial (overdispersed, no zero-inflation)
- If  $\pi_g > 0.3$  and  $\sigma_g^2/\mu_g \leq 1.5$ : use Zero-Inflated Poisson (equidispersed, with zero-inflation)
- If  $\pi_g > 0.3$  and  $\sigma_g^2/\mu_g > 1.5$ : use Zero-Inflated Negative Binomial (overdispersed, with zero-inflation)

For ZIP( $\lambda, \pi$ ) and ZINB( $\mu, \alpha, \pi$ ), where  $\alpha$  is the dispersion parameter. The theoretical moments are:

ZIP:

$$E[X] = (1 - \pi)\lambda, \quad \text{Var}(X) = (1 - \pi)(\lambda + \pi\lambda^2), \quad P(X = 0) = \pi + (1 - \pi)e^{-\lambda}. \quad (7)$$

ZINB:

$$E[X] = (1 - \pi)\mu, \quad \text{Var}(X) = (1 - \pi)(\mu + \alpha\mu^2 + \pi\mu^2), \quad P(X = 0) = \pi + (1 - \pi)(1 + \alpha\mu)^{-1/\alpha}. \quad (8)$$

Parameters for each selected model are then inferred from the target triplet  $(\hat{\mu}_g, \hat{\sigma}_g^2, \hat{\pi}_g)$  through moment matching and empirical optimization. Specifically, to estimate parameters, we minimize the log-scale squared error between theoretical and target parameters:

$$\min_{\theta} ( (\log_{10} E[X; \theta] - \log_{10} \hat{\mu}_g)^2 + (\log_{10} \text{Var}(X; \theta) - \log_{10} \hat{\sigma}_g^2)^2 + (\log_{10} P(X = 0; \theta) - \log_{10} \hat{\pi}_g)^2 ) \quad (16)$$

The log-scale objective is chosen because expression data spans multiple orders of magnitude, and relative errors are more meaningful than absolute errors in this context. The optimization is performed using scikit-learn’s numerical optimization routines.

For the simpler NB and Poisson models, parameters are derived directly from the target statistics without numerical optimization:

$$\text{NB: } \mu = \hat{\mu}_g, \quad \alpha = \max \left( 0, \frac{\hat{\sigma}_g^2 - \hat{\mu}_g}{\hat{\mu}_g^2} \right) \quad (17)$$

$$\text{Poisson: } \lambda = \hat{\mu}_g \quad (18)$$

The maximum operation in the NB dispersion parameter ensures numerical stability by preventing negative dispersion values.

**S1.5 Spatial Placement via Rank-based Assignment.** The resulting expression profiles are then spatially placed via a rank-based assignment procedure, which preserves local spatial ranks and co-expression module structure observed in the original slice. For each gene, the synthetic counts are sorted and assigned to spatial locations based on their relative ranks in the original slice’s spatial distribution. Specifically, if a gene’s highest-expressing spot in the original data is at location  $i$ , the corresponding high-count value in the synthetic data is also placed at an analogous spatial position in the synthetic slice. This rank-based approach ensures that spatial co-expression modules - groups of genes that are highly expressed in nearby regions - are preserved in the simulation while allowing the specific expression magnitudes to vary according to the fitted count distributions.

## S2 Gene Expression Alteration Details

**S2.1 Parametric Controls for Expression Modification.** To simulate biological conditions or disease-associated perturbations, FEAST provides a set of parametric controls for modifying gene expression characteristics in a controlled and interpretable way.

Mean fold change ( $c_\mu$ ):

$$\log \mu'_g = \log(\mu_g + \varepsilon) + \log c_\mu, \quad (19)$$

where  $\varepsilon$  is a small pseudocount (default:  $10^{-6}$ ) added to stabilize computation for genes with zero mean expression. A fold change of  $c_\mu = 2$  doubles the mean expression, while  $c_\mu = 0.5$  halves it. This control operates in log space to ensure consistent multiplicative scaling across the full expression range.

Dispersion scale ( $c_\phi$ ): the dispersion parameter  $\phi_g$  quantifies gene-specific overdispersion beyond the mean. It can be approximated from observed mean and variance statistics as:

$$\sigma_g^2 \approx \mu_g + \phi_g \mu_g^2, \quad \hat{\phi}_g = \max\left(0, \frac{\sigma_g^2 - \mu_g}{\mu_g^2}\right), \quad (20)$$

The adjusted variance under dispersion scaling is:

$$\sigma_g'^2 = \mu_g' + (c_\phi \hat{\phi}_g) \mu_g'^2, \quad (21)$$

here,  $c_\phi > 1$  increases variability (reflecting greater heterogeneity in expression), while  $c_\phi < 1$  suppresses variability. This control is particularly useful for simulating biological contexts in which gene expression becomes more or less noisy.

To model changes in zero-inflation (dropout rate) associated with different biological or disease-state perturbations, FEAST adopts a logit-space parameterization of the gene-specific dropout probability  $\pi_g$ :

$$\text{logit}(\pi'_g) = \delta_\pi + b_\pi \text{logit}(\pi_g) - \kappa \log \frac{\mu'_g}{\mu_g + \varepsilon}, \quad (22)$$

where:

- $\delta_\pi$  is an additive shift on the logit scale, controlling the baseline sparsity level;
- $b_\pi$  is a multiplicative slope on the original  $\text{logit}(\pi_g)$ , controlling how the gene-specific sparsity scales with its original value;
- $\kappa$  is a coupling parameter linking changes in sparsity to changes in mean expression; The term  $-\kappa \log(\mu'_g/\mu_g)$  ensures that, when mean expression increases and  $\kappa > 0$ , sparsity naturally decreases, consistent with typical biological behavior.

This three-parameter model provides flexible control over how dropout patterns shift under altered conditions, including both global shifts and expression-dependent adjustments.

## S3 Alignment Space Alteration

FEAST generates paired ST datasets with controlled geometric transformations, enabling rigorous benchmarking of spatial alignment and registration algorithms. The framework provides exact ground-truth spot-to-spot (cell-to-cell) correspondences while introducing realistic spatial distortions that mimic experimental variability such as tissue rotation, stretching, or local warping.

**S3.1 Workflow Overview.** The alignment simulation workflow consists of three main stages: (a) Expression simulation: a reference slice is first generated using the core FEAST expression simulation module, producing realistic gene expression and spatial distributions. (b) Geometric transformation: controlled spatial operations, including rotation, affine transformation, elastic deformation, or combined transformations, are applied to create a perturbed counterpart of the reference slice. These transformations are parameterized to preserve overall topology while introducing spatial distortions of adjustable intensity. (c) Ground truth construction: the exact mapping between corresponding spots in the reference and transformed slices is recorded as an alignment matrix, providing explicit ground truth for evaluating spatial registration and correspondence algorithms.

**S3.2 Rotation Transformations.** FEAST provides two distinct rotation schemes tailored to different ST technologies:

*Imaging-based rotation.* For array- or imaging-based platform with fixed spatial coordinates (e.g., 10x Visium, Slide-seq), rotation is applied directly to the coordinate matrix:

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad (23)$$

$$\mathbf{s}'_i = \mathbf{R}(\theta)(\mathbf{s}_i + \mathbf{c}) - \mathbf{c}, \quad (24)$$

where  $\mathbf{s}_i$  denotes the original coordinates,  $\mathbf{c}$  is the rotation center, and  $\theta$  is the rotation angle in radians. To preserve realistic tissue geometry, an optional filtering step retains only spots within the original spatial boundaries:

$$\mathcal{S}_{\text{filtered}} = \{\mathbf{s}'_i : x_{\min} \leq s'_{i,x} \leq x_{\max}, y_{\min} \leq s'_{i,y} \leq y_{\max}\}, \quad (25)$$

where  $(x_{\min}, x_{\max}, y_{\min}, y_{\max})$  define bounding box of the original tissue section.

*Sequencing-based rotation.* For sequencing-based platforms with irregular spot distributions, FEAST implements a grid-realignment strategy to simulate the discrete capture-array structure:

1. Grid generation: a hexagonal grid is constructed to approximate the original spot density. The minimal inter-spot distance,  $d_{\min}$ , is estimated using k-nearest-neighbors algorithm. Then:

$$s = \frac{d_{\min}}{2}, \quad h = \frac{d_{\min}\sqrt{3}}{2}, \quad (26)$$

where  $s$  is horizontal spacing and  $h$  is vertical offset. The Grid coordinates are:

$$\mathbf{g}_{r,c} = [x_{\min} + 2cs + (r \bmod 2)s, y_{\min} + rh], \quad r \in [0, R), c \in [0, C), \quad (27)$$

with  $R = \lceil (y_{\max} - y_{\min})/h \rceil + 1$  rows and  $C = \lceil (x_{\max} - x_{\min})/(2s) \rceil + 1$  columns.

2. Rotation and assignment: Each original spot coordinate  $\mathbf{s}_i$  is rotated using the same rotation matrix as in the imaging-based method to obtain  $\mathbf{s}'_i$ . The rotated spot is then assigned to its nearest grid point using k-d tree search:

$$\mathbf{s}''_i = \arg \min_{\mathbf{g} \in \mathcal{G}} \|\mathbf{s}'_i - \mathbf{g}\|_2, \quad (28)$$

where  $\mathcal{G}$  is the set of all grid points.

3. Collision resolution: if multiple spots map to the same grid coordinate, only the first assignment is retained, ensuring that the capture layout remains discrete and non-overlapping.

The alignment ground-truth matrix is directly derived from the original cell/spot indices, ensuring that each rotated spot is mapped back to its corresponding original spot.

**S3.3 Edge Artifact Filtering.** Geometric transformations often introduce edge artifacts where rotated tissue extends beyond original boundaries. FEAST provides optional edge filtering to remove spots near boundaries:

$$\mathcal{S}_{\text{clean}} = \{\mathbf{s}'_i : x_{\min} + \delta_x < s'_{i,x} < x_{\max} - \delta_x, y_{\min} + \delta_y < s'_{i,y} < y_{\max} - \delta_y\}, \quad (29)$$

where  $\delta_x = \rho(x_{\max} - x_{\min})$  and  $\delta_y = \rho(y_{\max} - y_{\min})$  with margin ratio  $\rho$  (default: 0.03 or 3%).



## S4 Spot-Resolution Simulation Details

From single cells resolution slice  $\mathbf{Y}_{\text{sc}}$  with annotations  $z_n \in \{1, \dots, K\}$ , FEAST synthesizes multi-resolution mixtures with ground-truth compositions. After the simulation of single slice, we could conduct aggregation.

**Multi-resolution aggregation.** Given a fine level  $(\mathbf{X}^{(h)}, W^{(h)}, \mathbf{s}^{(h)})$  and aggregation matrix  $\mathbf{A}$ ,

$$\mathbf{X}^{(\ell)} = \mathbf{A}\mathbf{X}^{(h)}, \quad L^{(\ell)} = \mathbf{A}L^{(h)}, \quad W_{ik}^{(\ell)} = \frac{\sum_u A_{iu} L_u^{(h)} W_{uk}^{(h)}}{\sum_u A_{iu} L_u^{(h)}}, \quad (30)$$

preserving mass and proportions across resolutions.

## S5 Benchmark Metrics

To rigorously evaluate the performance of spatial transcriptomics methods, we established a comprehensive benchmarking framework that covers four primary tasks: simulation, clustering, alignment, and interpolation. For each task, a suite of quantitative metrics has been designed to assess various facets of algorithmic performance.

**S5.1 Simulation Benchmark Metrics.** These metrics assess the fidelity of simulated spatial transcriptomics data compared to the original data it was modeled from.

To evaluate Distributional Fidelity:

- Mean and Variance Correlation: the Pearson correlation between the gene-wise mean and variance statistics across all genes in the real ( $X_{\text{real}}$ ) and simulated ( $X_{\text{sim}}$ ) data quantifies the similarity of marginal expression distributions.
- Gene and Spot Zero Proportion KS Tests: the Kolmogorov-Smirnov (KS) test is applied to assess the similarity between the distributions of zero proportions calculated per-gene and per-spot in the real and simulated datasets.

Formally, let  $\mu_{\text{real},j}$  and  $\mu_{\text{sim},j}$  denote the mean expression of gene  $j$  in real and simulated data respectively, and  $v_{\text{real},j}, v_{\text{sim},j}$  denote their corresponding variances across spots. The Pearson correlations are then computed as:

$$r_{\text{mean}} = \text{corr}((\mu_{\text{real},j})_{j=1}^g, (\mu_{\text{sim},j})_{j=1}^g), \quad r_{\text{var}} = \text{corr}((v_{\text{real},j})_{j=1}^g, (v_{\text{sim},j})_{j=1}^g). \quad (31)$$

For the KS test comparing zero-proportion distributions (either per-gene or per-spot), let  $F_1(x)$  and  $F_2(x)$  denote the empirical cumulative distribution functions (CDFs) of the two groups, the KS statistic is defined as:

$$D = \sup_x |F_1(x) - F_2(x)|. \quad (32)$$

- Relative Error of Mean Expression: the average relative error between the mean expression of each gene in the real and simulated data.

$$\text{RE}_{\text{mean}} = \frac{1}{g} \sum_{j=1}^g \frac{|\mu_{\text{real},j} - \mu_{\text{sim},j}|}{\mu_{\text{real},j} + \epsilon} \quad (33)$$

and  $\epsilon$  means a small constant to avoid denominator to be zero.

- Spatial Autocorrelation: Moran's I and Geary's C are computed for both real and simulated data to ensure spatial patterns are preserved.

Given a spatial weights matrix  $W = (w_{ij})$  with total weight  $\mathcal{W} = \sum_i \sum_j w_{ij}$ , let  $x_i$  be the expression (or a summary) at spot  $i$ ,  $\bar{x} = n^{-1} \sum_i x_i$ , and standardized values  $z_i = x_i - \bar{x}$ . Moran's I and Geary's C are

$$I_{\text{Moran}} = \frac{n}{\mathcal{W}} \cdot \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}, \quad (34)$$

$$C_{\text{Geary}} = \frac{(n-1)}{2\mathcal{W}} \cdot \frac{\sum_i \sum_j w_{ij} (x_i - x_j)^2}{\sum_i (x_i - \bar{x})^2}. \quad (35)$$



**S5.2 Clustering Benchmark Metrics.** These metrics assess the quality of spatial clustering results.

**Spatial coherence of clusters.** These metrics evaluate the spatial organization and continuity of the identified clusters.

- CHAOS (CHoice of Anchor points in the Original Space): this metric measures the spatial disorganization of clusters by calculating the average distance from each spot to its nearest neighbor within the same cluster. Lower values suggest more compact and well-organized clusters. After standardizing the spatial coordinates  $S$ , it is computed as:

$$\text{CHAOS} = \frac{1}{n} \sum_{k \in K} \sum_{i: c_i = k} \min_{j \neq i, c_j = k} d(s'_i, s'_j) \quad (36)$$

where  $K$  is the set of clusters and  $d(\cdot, \cdot)$  is the Euclidean distance.

- PAS (Proportion of Adversarial Spots): this metric quantifies the spatial continuity by identifying the proportion of spots whose local neighborhood is dominated by other clusters. A lower PAS indicates better spatial continuity.

$$\text{PAS} = \frac{1}{n} \sum_{i=1}^n \mathbb{I} \left[ \sum_{j \in \mathcal{N}_k(i)} \mathbb{I}[c_j \neq c_i] > \frac{k}{2} \right] \quad (37)$$

where  $\mathcal{N}_k(i)$  is the set of  $k$ -nearest neighbors of spot  $i$  (default  $k = 10$ ).

**Standard clustering evaluation.** When ground-truth labels are available, we employ widely-used metrics to compare the predicted labels ( $\hat{Y}$ ) with the true labels ( $Y$ ):

- Adjusted Rand Index (ARI)
- Normalized Mutual Information (NMI)

Let  $n$  denote the number of spots, and  $n_{ij}$  represent the entries of the contingency table between true clusters  $i$  and predicted clusters  $j$ . Define row sums  $a_i = \sum_j n_{ij}$  and column sums  $b_j = \sum_i n_{ij}$ . Using  $\binom{u}{2} = u(u-1)/2$ , the ARI is computed as:

$$\text{ARI} = \frac{\sum_{i,j} \binom{n_{ij}}{2} - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}}{\frac{1}{2} \left( \sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right) - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}} \quad (38)$$

For NMI, given the mutual information  $I(Y; \hat{Y})$  and entropies  $H(Y), H(\hat{Y})$ , we use symmetric normalization as:

$$\text{NMI} = \frac{I(Y; \hat{Y})}{\sqrt{H(Y) H(\hat{Y})}}, \quad I(Y; \hat{Y}) = \sum_{i,j} \frac{n_{ij}}{n} \log \frac{n_{ij}}{a_i b_j} \quad (39)$$

**S5.3 Alignment Benchmark Metrics.** These metrics evaluate the quality of the spatial alignment between two tissue slices.

**Matching accuracy.** Given a predicted alignment probability matrix  $\Pi \in \mathbb{R}^{n_1 \times n_2}$  and a ground-truth matching matrix  $G \in \{0, 1\}^{n_1 \times n_2}$ , where  $n_1$  and  $n_2$  are the number of spots in each slice, we first derive a binary prediction matrix  $P$  by selecting the most likely match for each spot in the first slice:

$$P_{ij} = \begin{cases} 1 & \text{if } j = \arg \max_k \Pi_{ik} \\ 0 & \text{otherwise} \end{cases} \quad (40)$$

From this, we compute the following based on the confusion matrix:

- True Positives (TP):  $\sum_{i,j} P_{ij} \cdot G_{ij}$

- False Positives (FP):  $\sum_{i,j} P_{ij} \cdot (1 - G_{ij})$
- False Negatives (FN):  $\sum_{i,j} (1 - P_{ij}) \cdot G_{ij}$

The performance is then quantified using:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (41)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (42)$$

$$\text{F1 Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (43)$$

**Alignment consistency.** This metric assesses the bidirectionality of the alignment. A consistent alignment should have the forward mapping from slice 1 to slice 2 correspond with the backward mapping from slice 2 to slice 1.

The forward and backward mappings are defined as:

$$f(i) = \arg \max_j \Pi_{ij} \quad (\text{forward: slice 1} \rightarrow \text{slice 2}) \quad (44)$$

$$b(j) = \arg \max_i \Pi_{ij} \quad (\text{backward: slice 2} \rightarrow \text{slice 1}) \quad (45)$$

Consistency is the fraction of spots in slice 1 that are mapped back to themselves:

$$\text{Consistency} = \frac{1}{n_1} \sum_{i=1}^{n_1} \mathbb{I}[b(f(i)) = i] \quad (46)$$

where  $\mathbb{I}[\cdot]$  is the indicator function.

**Region and gene expression concordance.** These metrics evaluate how well the alignment preserves biological structures and expression patterns.

- Region Accuracy: it measures the proportion of correctly aligned spots that map to the same annotated region. Given region labels  $R_1$  and  $R_2$  for the two slices and a ground-truth mask  $M$  indicating which spots have a valid match:

$$\text{Region Accuracy} = \frac{\sum_{i: M_i=1} \mathbb{I}[r_1^{(i)} = r_2^{(f(i))}]}{\sum_i M_i} \quad (47)$$

- Gene Expression Correlation: for successfully aligned spot pairs, this metric computes the average Pearson correlation of their gene expression profiles. Given expression matrices  $X_1, X_2 \in \mathbb{R}^{n \times g}$  for  $n$  aligned spots and  $g$  common genes:

$$\rho_i = \frac{\sum_{k=1}^g (X_{1,ik} - \bar{X}_{1,i})(X_{2,f(i),k} - \bar{X}_{2,f(i)})}{\sqrt{\sum_{k=1}^g (X_{1,ik} - \bar{X}_{1,i})^2} \sqrt{\sum_{k=1}^g (X_{2,f(i),k} - \bar{X}_{2,f(i)})^2}} \quad (48)$$

The overall correlation is the average over all aligned spots:  $\frac{1}{n} \sum_{i=1}^n \rho_i$ .

**S5.4 Interpolation Benchmark Metrics.** These metrics evaluate the quality of interpolated gene expression data by comparing it to a ground-truth slice.

#### **Statistical similarity.**

- Mean and Variance Correlation: it computes the Pearson correlation between the gene-wise means and variances of the real ( $X_{\text{real}}$ ) and predicted ( $X_{\text{pred}}$ ) data.
- Zero Proportion KS Test: the Kolmogorov-Smirnov (KS) test is used to compare the distributions of the proportion of zero-expression values per gene in the real and predicted data.

The same definitions of the Pearson correlation and KS statistic as described in Section S5.1 are applied here to compare  $X_{\text{real}}$  versus  $X_{\text{pred}}$ , assessing correspondence in gene-wise means and variances as well as zero-proportion distributions.

### ***Structural and perceptual similarity.***

- Structural Similarity (Moran’s I Correlation): this measures how well the spatial structure of gene expression is preserved by calculating the Pearson correlation between the Moran’s I values of each gene in the real and predicted data.
- Perceptual Hash Similarity:<sup>34</sup> this metrics is derived from image similarity metrics. For each gene, we generate a binary “hash” of its spatial expression pattern by downsampling to a small grid (e.g., 8x8) and thresholding at the median expression. The similarity is the average Hamming distance between the hashes of the real and predicted data across all genes. This metric could be reliable and robust to compare the overall spatial similarity, even though the two slices are not in exactly same shape.

For structural similarity, we reuse the Pearson correlation (as in S5.1) computed between per-gene Moran’s I values from real and predicted data. For perceptual hashes, let  $h_{\text{real}}^{(g)}, h_{\text{pred}}^{(g)} \in \{0, 1\}^m$  be the  $m$ -bit hashes for gene  $g$  from real and predicted slices. With Hamming distance  $d_H$ , define the similarity as

$$S_{\text{pHash}} = 1 - \frac{1}{G} \sum_{g=1}^G \frac{d_H(h_{\text{real}}^{(g)}, h_{\text{pred}}^{(g)})}{m}. \quad (49)$$

### **S6 Data Availability**

Dataset 1 comprises 12 human DLPFC sections from the spatialLIBD resource (<http://research.libd.org/spatialLIBD/>); DLPFC 151670 was used for simulator benchmarking, 151676 for clustering evaluation, and 151675 for alignment testing. Dataset 2 consists of 3D MERFISH spatial transcriptomics data with 147 annotated coronal sections from the Allen Brain Atlas ([https://alleninstitute.github.io/abc\\_atlas\\_access/descriptions/Zhuang-ABCA-1.html](https://alleninstitute.github.io/abc_atlas_access/descriptions/Zhuang-ABCA-1.html)); sections ID 6 and 7 were used for simulator benchmarking, section ID 7 for deconvolution testing, and sections ID 5-9 for interpolation evaluation. Dataset 3 contains 19 consecutive slices of annotated 3D OpenST sequencing data from human lymph node (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE251926>); slices ID 5 and 6 were used for simulator benchmarking. Dataset 4 includes five annotated Stereo-seq slices from the MOSTA mouse atlas (<https://www.sciencedirect.com/science/article/pii/S0092867422003993>); sample E14\_5\_E2S2 was used for simulator benchmarking. Dataset 5 contains one Slide-seqV2 slice obtained from the Tencent SpatialOmics database (<https://gene.ai.tencent.com/SpatialOmics/dataset?datasetID=119>). Dataset 6 includes Human Lymph Node Preview Data sequenced with Xenium from 10x Genomics public datasets (<https://www.10xgenomics.com/datasets/human-lymph-node-preview-data-xenium-human-multi-tissue-and-cancer-panel-1-standard>) and was used for simulator benchmarking. Dataset 7 includes two MERFISH spinal cord slices from [https://github.com/shanmeltzer/Spinal\\_MERFISH](https://github.com/shanmeltzer/Spinal_MERFISH).